

1.1.2 Databases: Dynamic sites need data

Views are key to the *dynamic* part of a dynamic website. They can create what is on a page based on data or input from the user. That data might be products in a catalog, the score of last night's game, or your latest recipe for gnocchi. In all these cases, your data has to live somewhere. Enter the database.

Databases are a field in computing in their own right. There are different kinds, each with its advantages and disadvantages. There are many implementations of each kind and specialty languages written just for communicating with them. All of this can be a little overwhelming. Django implements an ORM to abstract this away.

Consider a database containing information about books, like the one shown in figure 1.6. You are going to have different kinds of data: an author (that's me), a publisher (that's Manning), and a book (that's what you're holding). There are many authors, many books, and many publishers. Some books have multiple authors, and some authors have written multiple books. A relational database stores this information in tables similar to a sheet in an Excel spreadsheet. It then expresses relationships between these items with keys. The Author table might reference the Book table using a key that is a unique identifier for a Book.

Item tables

Book table

ID	Title	Year
10	Martian Chronicles, The	1950
11	Fahrenheit 451	1953
12	Carrie	1974
13	Firestarter	1980
14	Talisman, The	1984
15	Mystery	1993

Unique key
for each item

Author table

ID	Last_name	First_name
1	Bradbury	Ray
2	King	Stephen
3	Straub	Peter

Publisher table

ID	Name
23	Viking Press
24	Doubleday
25	Ballantine
26	Dutton

Relationship tables

Book-publisher table

ID	Book_ID	Publisher_ID
40	10	24
41	11	25
42	12	24
43	13	23
44	14	23
45	15	26

Doubleday published
both the Martian
Chronicles and Carrie.

Keys relate Books
to Publishers.

Book-author table

ID	Book_ID	Author_ID
80	10	1
81	11	1
82	12	2
83	13	2
84	14	2
85	14	3
86	15	3

Two Authors,
one Book

Keys relate Books
to Authors.

Figure 1.6 Tables store either item information or relationship information